

ECE 50877 Introduction to Applied Cryptography Project Proposals

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2 Differential Timing Analysis for Cryptographic Library Fingerprinting

This proposal originates with Dr. Zahra Ghodsi of Purdue University.

Title: Differential Timing Analysis for Cryptographic Library Fingerprinting

Problem Definition: The starting step of exploitation is determining the target's technology stack. An important part of this process is identifying the cryptographic libraries and implementations running on the target. Existing tools can often identify software through explicit version information or protocol behavior, but such information may be unavailable or intentionally hidden. This motivates the following research question: Can differential timing analysis identify the cryptographic library and version used by a remote target, despite network and system noise?

Proposed Methodology: We will construct a testbed containing multiple implementations and versions of commonly used cryptographic libraries. We will first collect timing measurements while each implementation performs a controlled suite of cryptographic operations under different inputs and configurations. We will analyze the resulting timing distributions to identify features that distinguish implementations. We will then train and evaluate statistical and machine-learning classifiers using these timing features.

Expected Findings: We expect the number of samples required to train accurate classifiers to rise substationally with noise. We also expect machine learning classifiers to outperform statistical classifiers, indicating broader applicability of machine learning techniques to side channel analysis.